Data Analysis Associate

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Comparison Analysis

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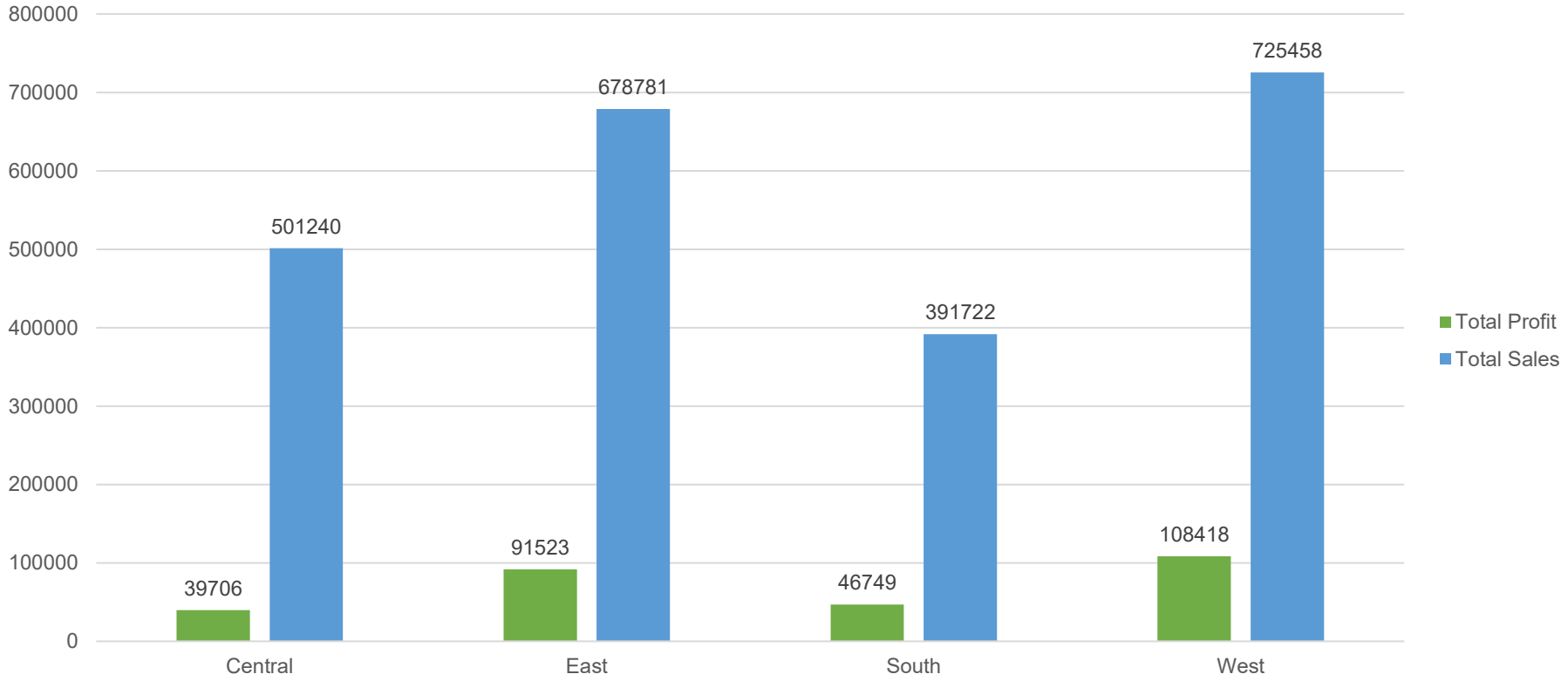
Comparison Analysis

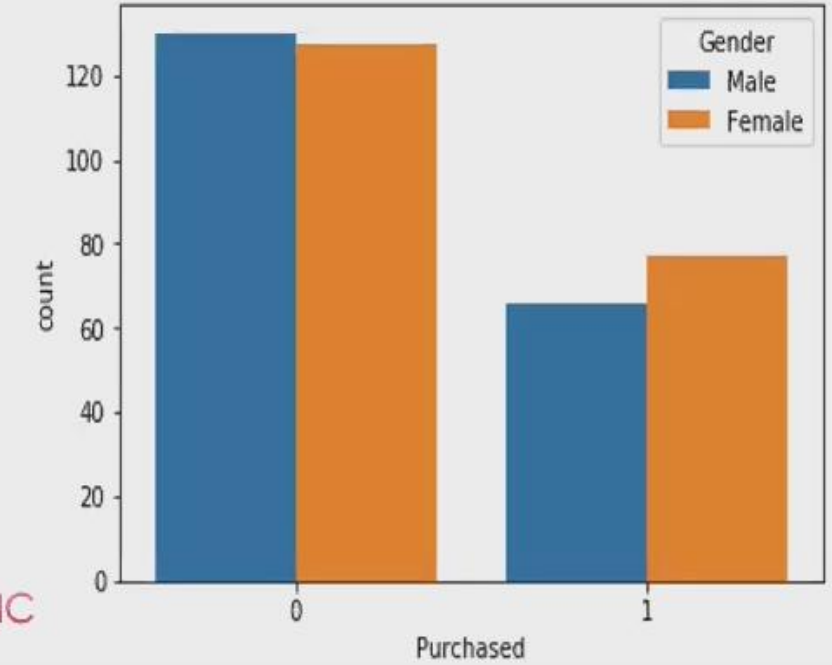
Comparison analysis involves comparing different variables, groups, or scenarios to identify similarities, differences, patterns, and trends.

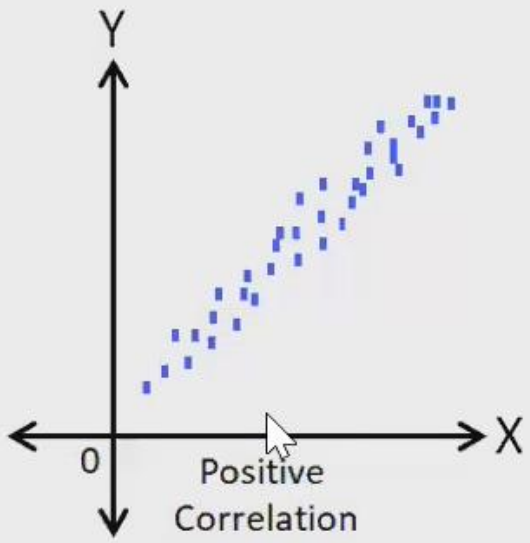
Eg. Compare sales and profit across all the regions.

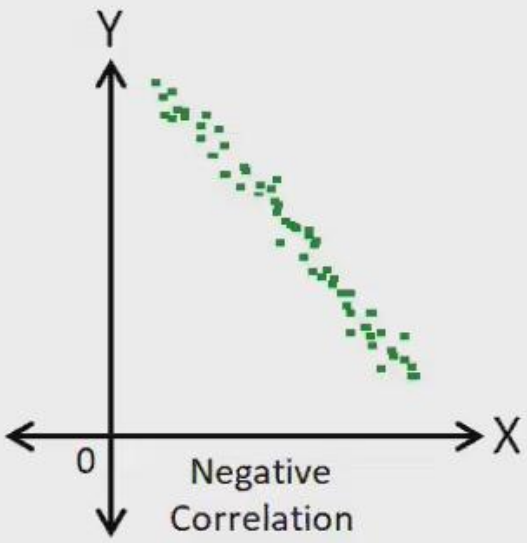


Comparison Vs Correlation

COMPARISON	CORRELATION
To compare different entities, variables or groups.	Examines the strength and direction of linear relationship between two continuous features
Tells the relative performance, characteristics or behaviour of different variables.	Quantifies the degree of association or dependency between two variables, indicating how change in one variable relates to change in another variable.







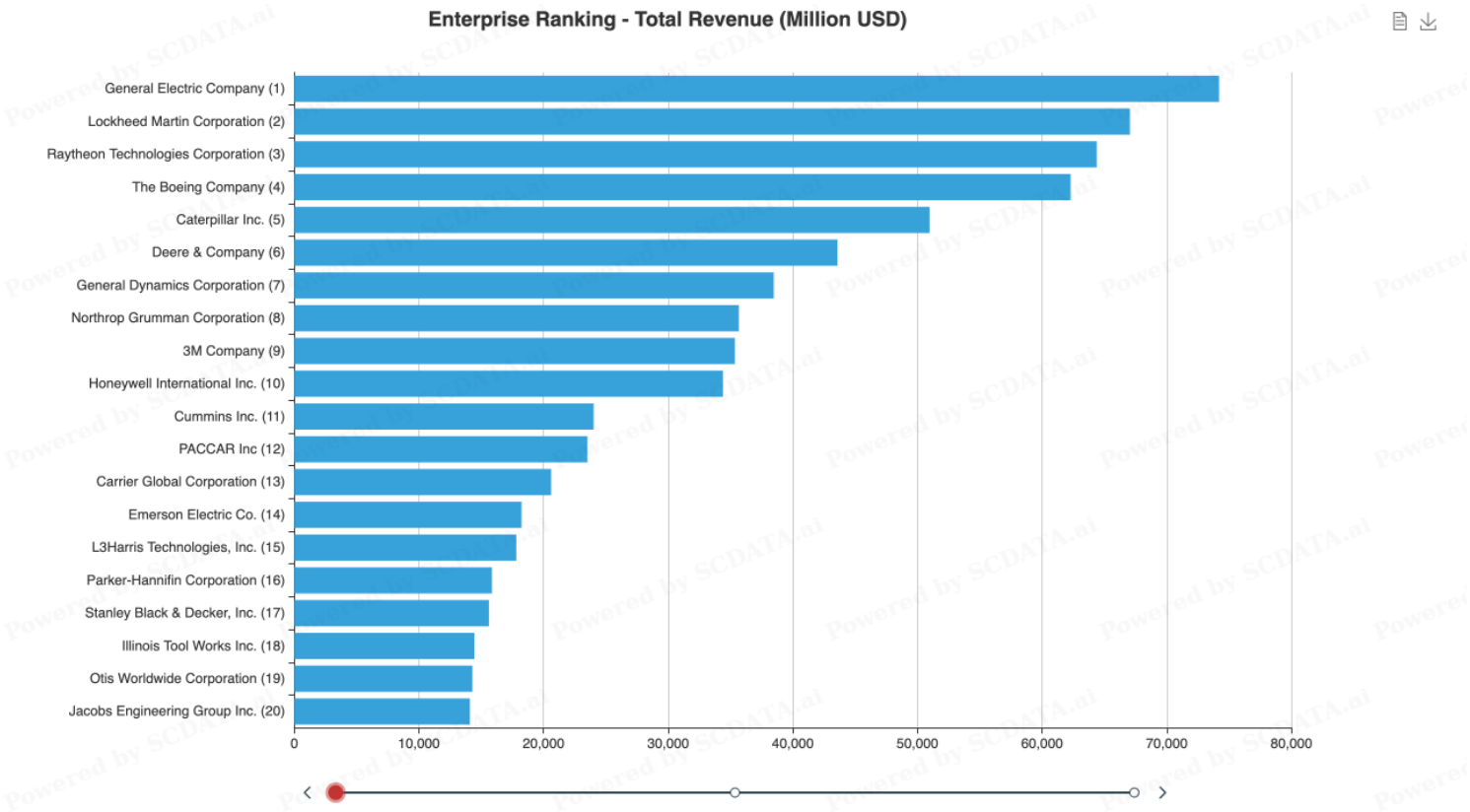
SKILLOGIC

COMPARISON	CORRELATION
Involves descriptive statistics, visualizations or hypothesis testing techniques.	These coefficients ranges from -1 to 1.
Common methods like T-test, ANOVA test, Chi-Square test are used	Pearson's correlation coefficient, Spearman's Rank correlation coefficient.

Ranking Analysis

Process of evaluating and analyzing the rankings of items, entities or data, based on certain criteria or attributes.

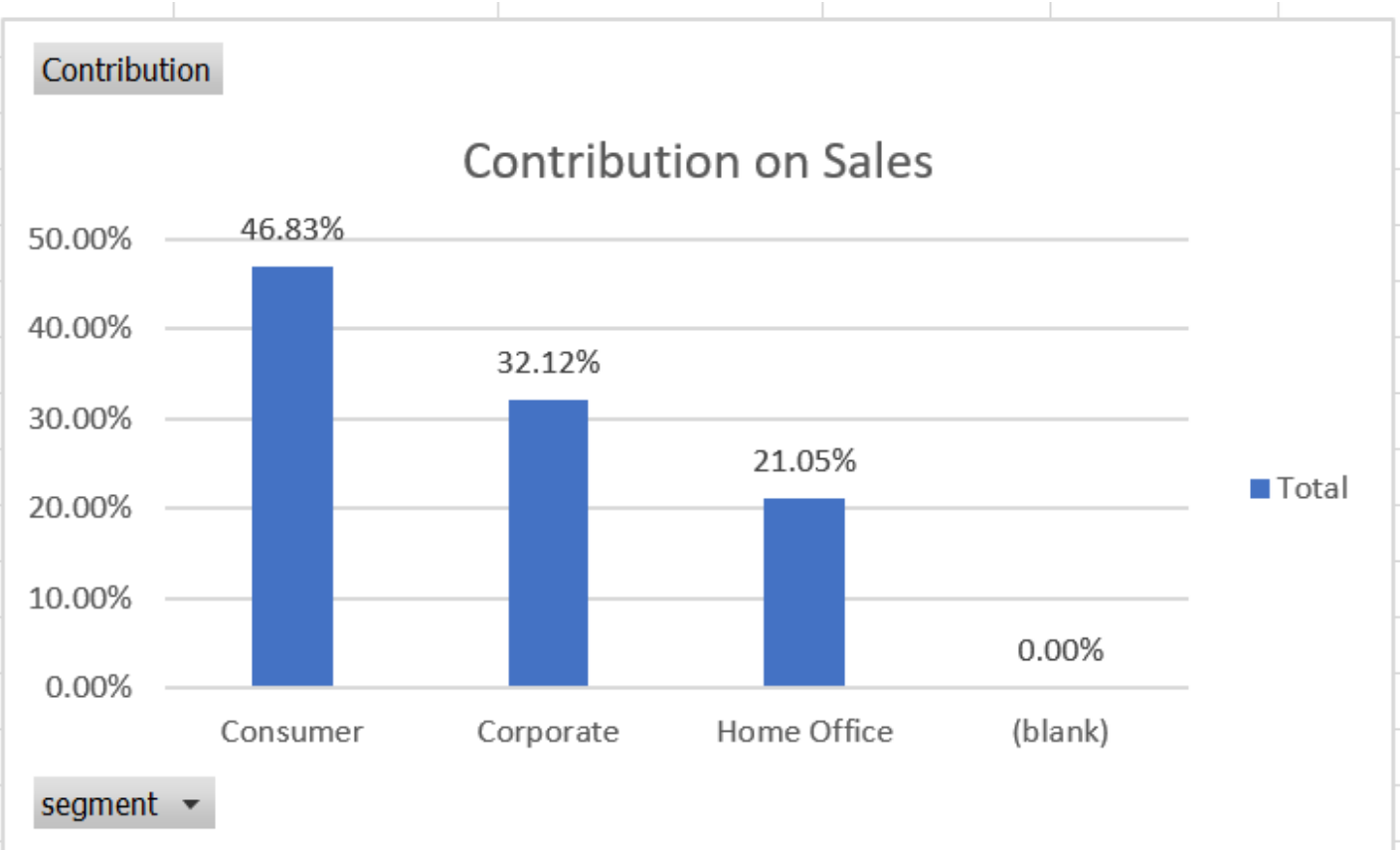
Eg. Find the top 5 products in terms of Sales.



Contribution Analysis

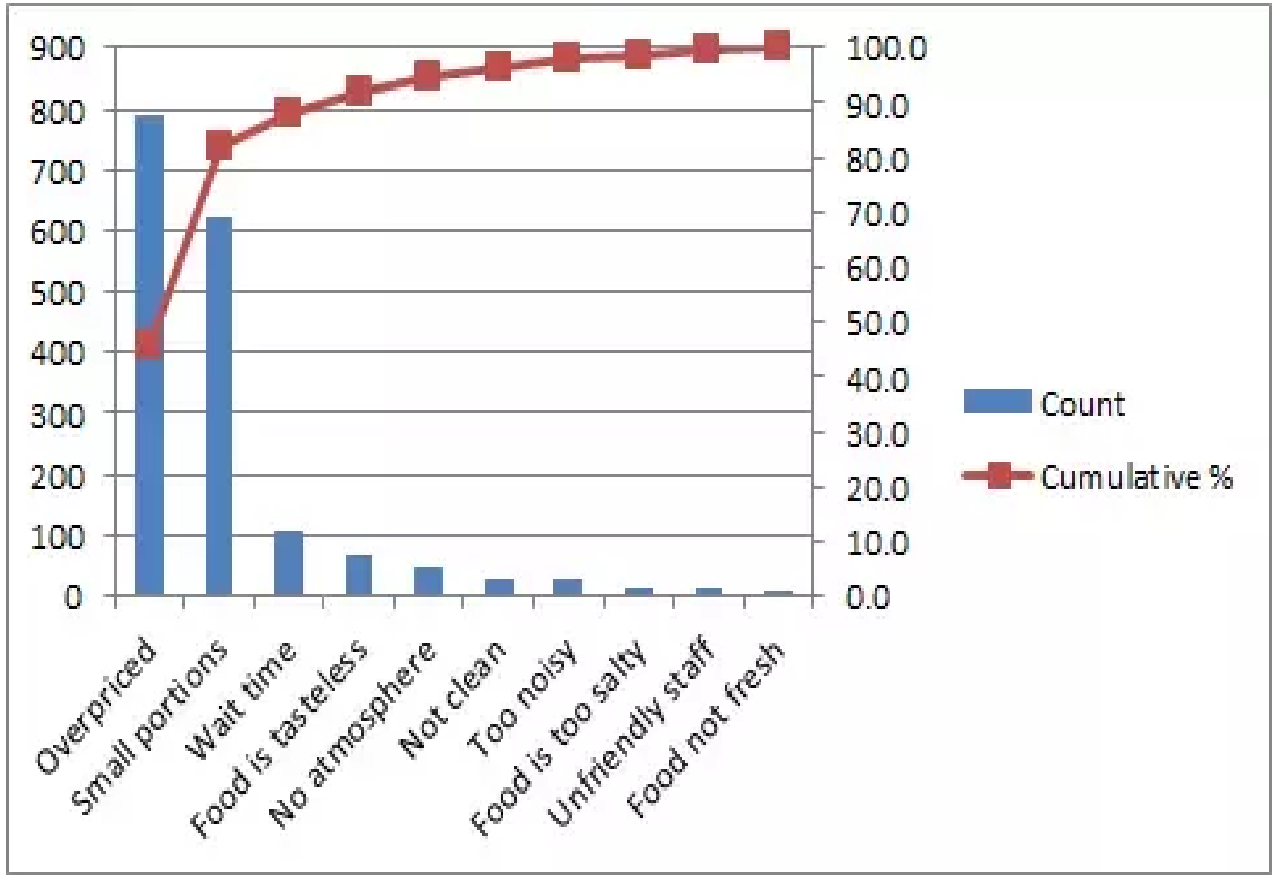
Contribution analysis is a powerful tool that helps businesses and organizations to analyze the contribution of various factors towards a particular outcome.

Eg. Analyze the contribution of categories towards Sales, Revenue, etc



Pareto Analysis

Pareto analysis, also known as the 80/20 rule, is a management tool used to identify the most significant factors or causes that contribute to a particular outcome. In Pareto analysis, it is assumed that 80% of the effects are caused by 20% of the factors.



Restaurant

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- Also called an 80-20 rule, 80% of the problems are caused by 20% of facts whether if you look into profits, customer complaints, technical issues, product defects, etc.
- It statistically separates a limited number of input factors that have a greater impact on an outcome
- It is a powerful quality and decision-making tool.
- It organizes and displays information to show the relative importance of various problems or causes of problems.

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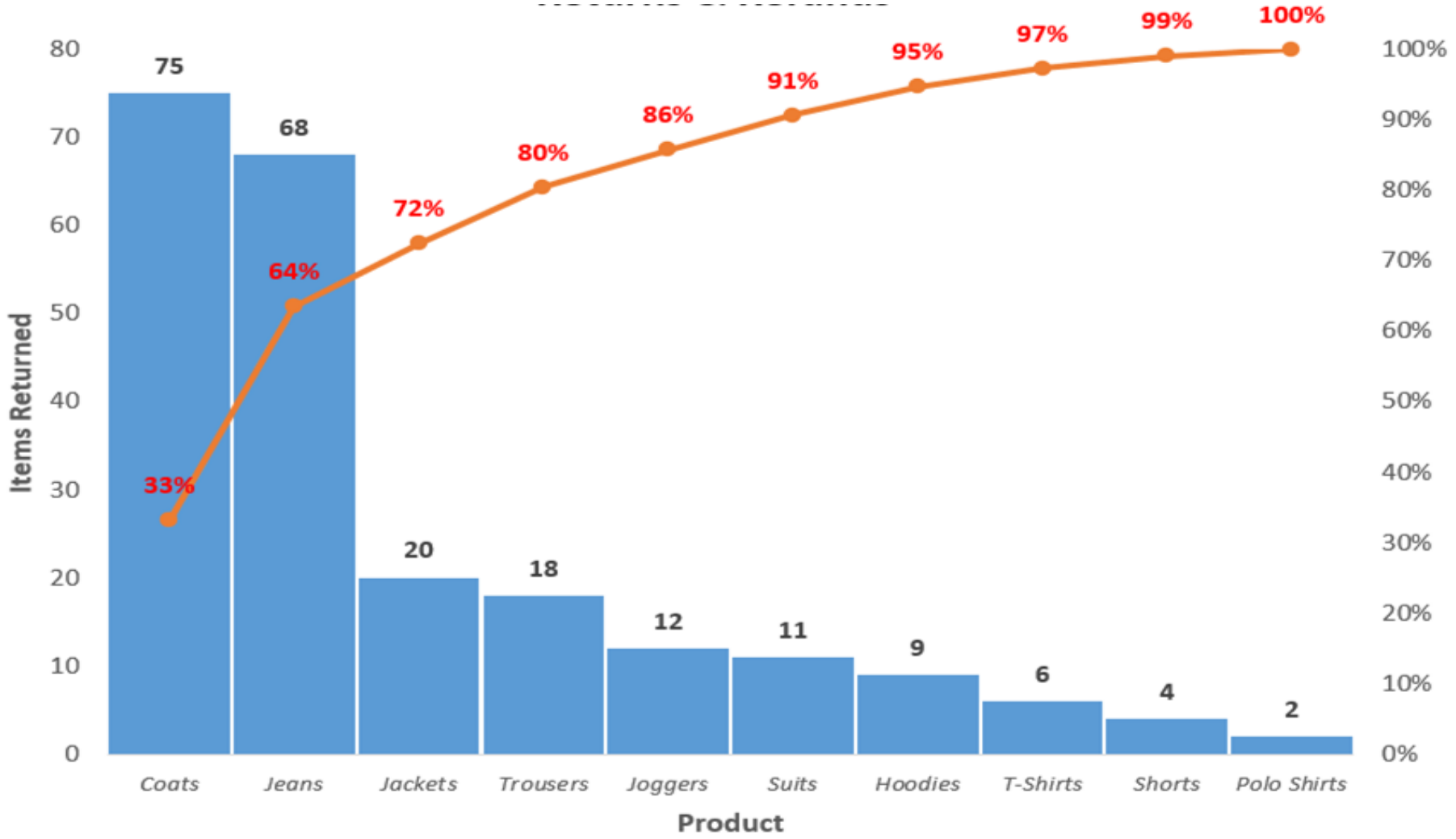
Returns & Refunds



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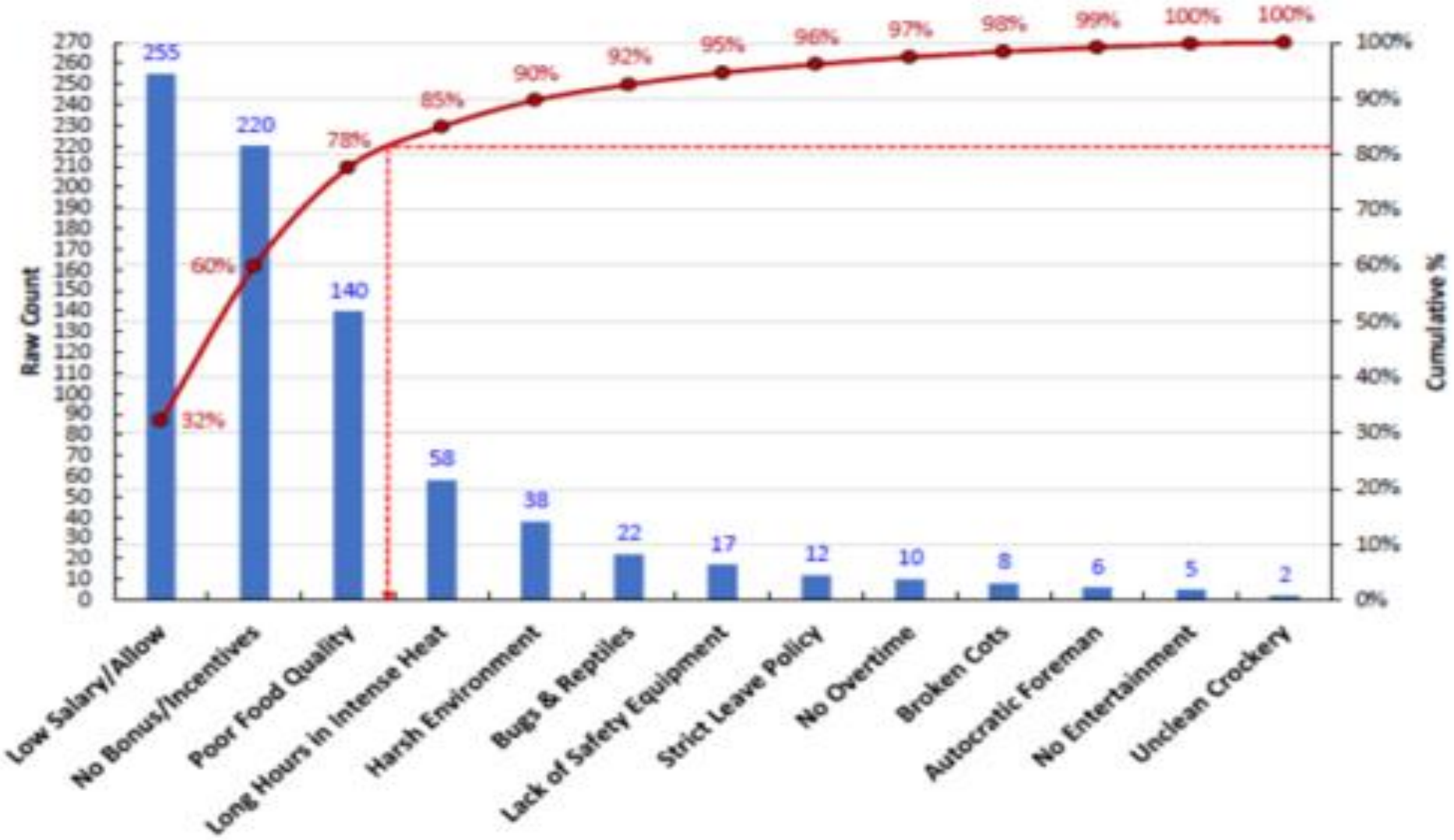
Attrition



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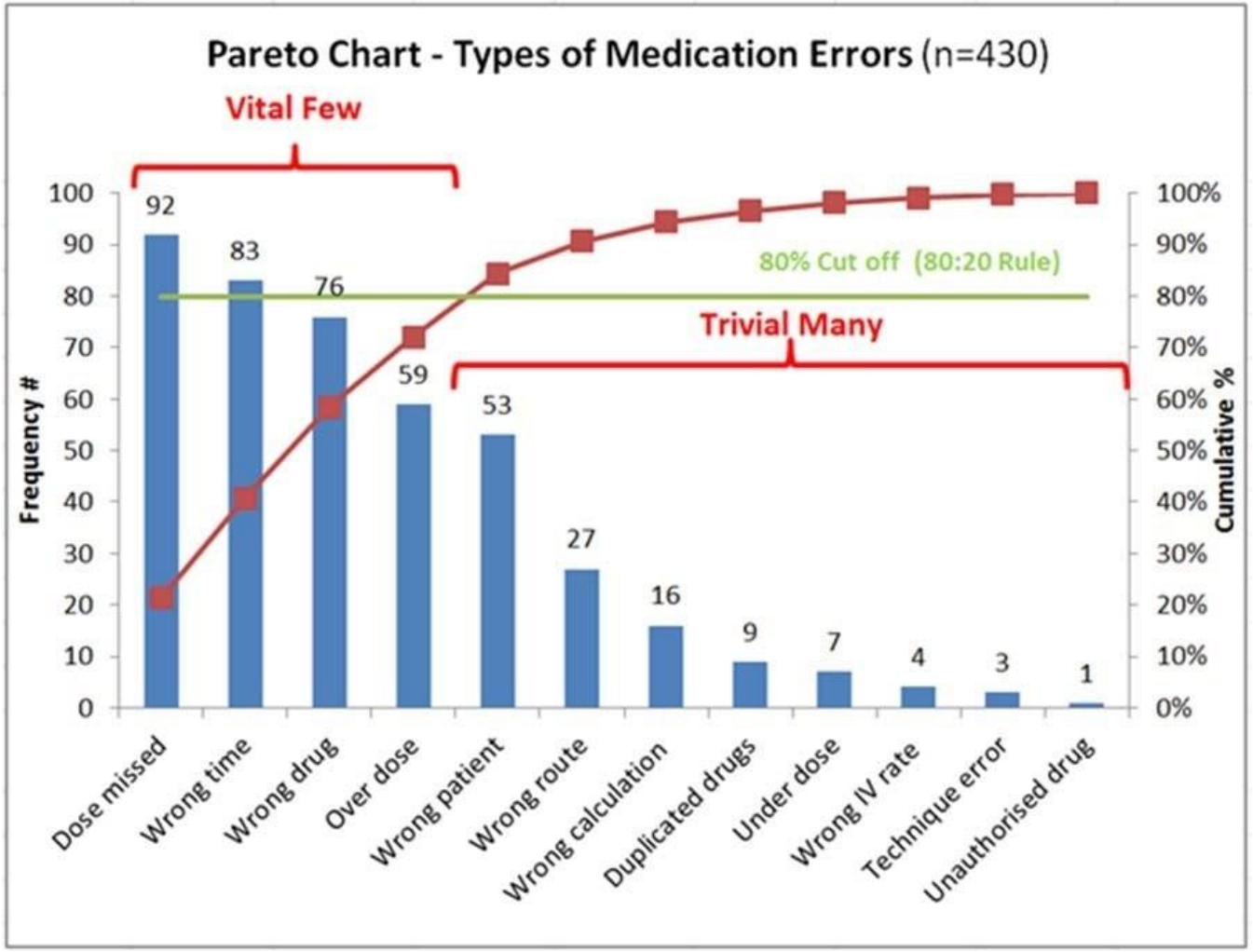
Medication Errors



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Interactive Dashboards

Interactive dashboards in Excel are a powerful way to present and analyze data. They allow users to explore and analyze data through interactive charts, tables, and visualizations.

It is a Data Management tool which comes with various interactive features where users can explore the data on a deeper level.

- Prepare your data.
- Create your dashboard with charts, tables or visualizations.
- Add interactivity using slicer, filters, pivot table etc
- Customize your dashboard

With interactivity, you can create a powerful and effective dashboard that will help you make better decisions based on your data.

Time Series Data

- Time series data is a sequence of data points that measure some variable over ordered period of time. It helps to understand and forecast data patterns. It is important to look for the pattern or components in a Time Series data such as Trend, Seasonality etc.
- Examples: Weather Forecasting, Sales Trends, Data captured in CCTV, Stock price, etc

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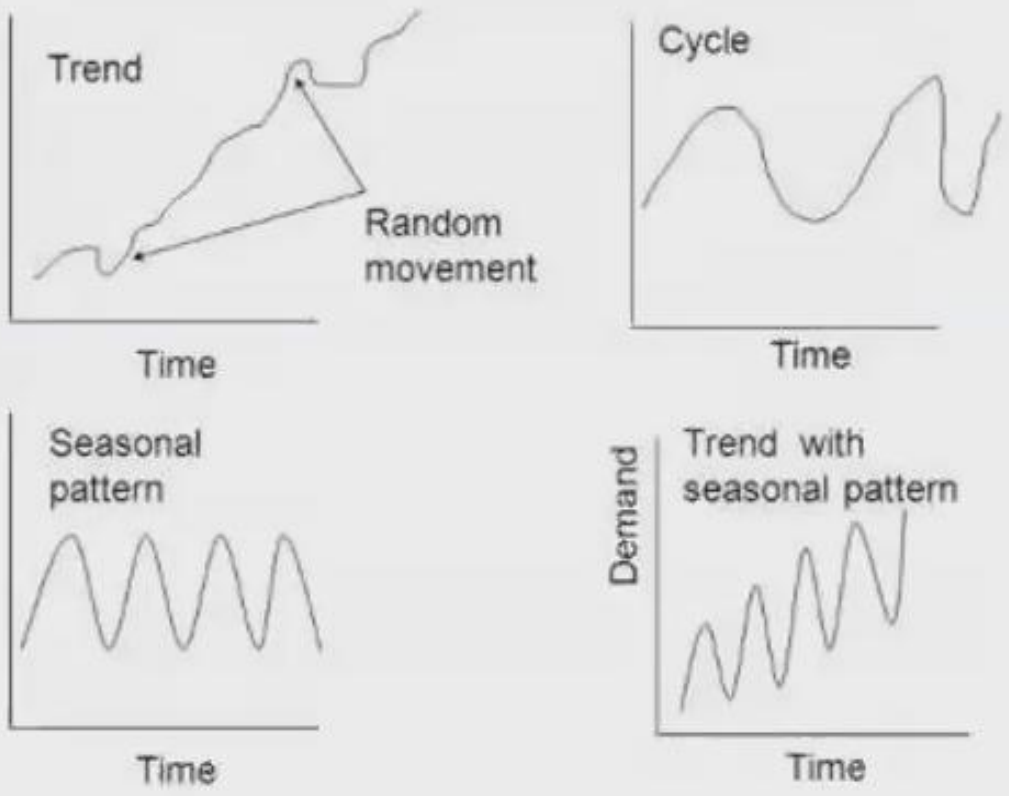
Patterns in Time Series

Trend: a long-term movement in the data, showing a general direction of the time series, either upwards or downwards, over a significant period.

Seasonality: It refers to regular, predictable changes in the data that repeat over a fixed period, such as monthly, quarterly, or yearly patterns.

Cyclic: These are the fluctuations in the data occurring at irregular intervals, typically influenced by economic or business cycles, with durations longer than a year.



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Trend Analysis

- **Trend** is a pattern in data that shows the movement of a series to relatively higher or lower values over a long period of time. In other words, a trend is observed when there is an increasing or decreasing slope in the time series.

- Hands-on – Trend Analysis for a Time Series Data.

Line Chart:

- Show how a continuous variable changes over time.

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Types of Trend

- **Upward trend:**

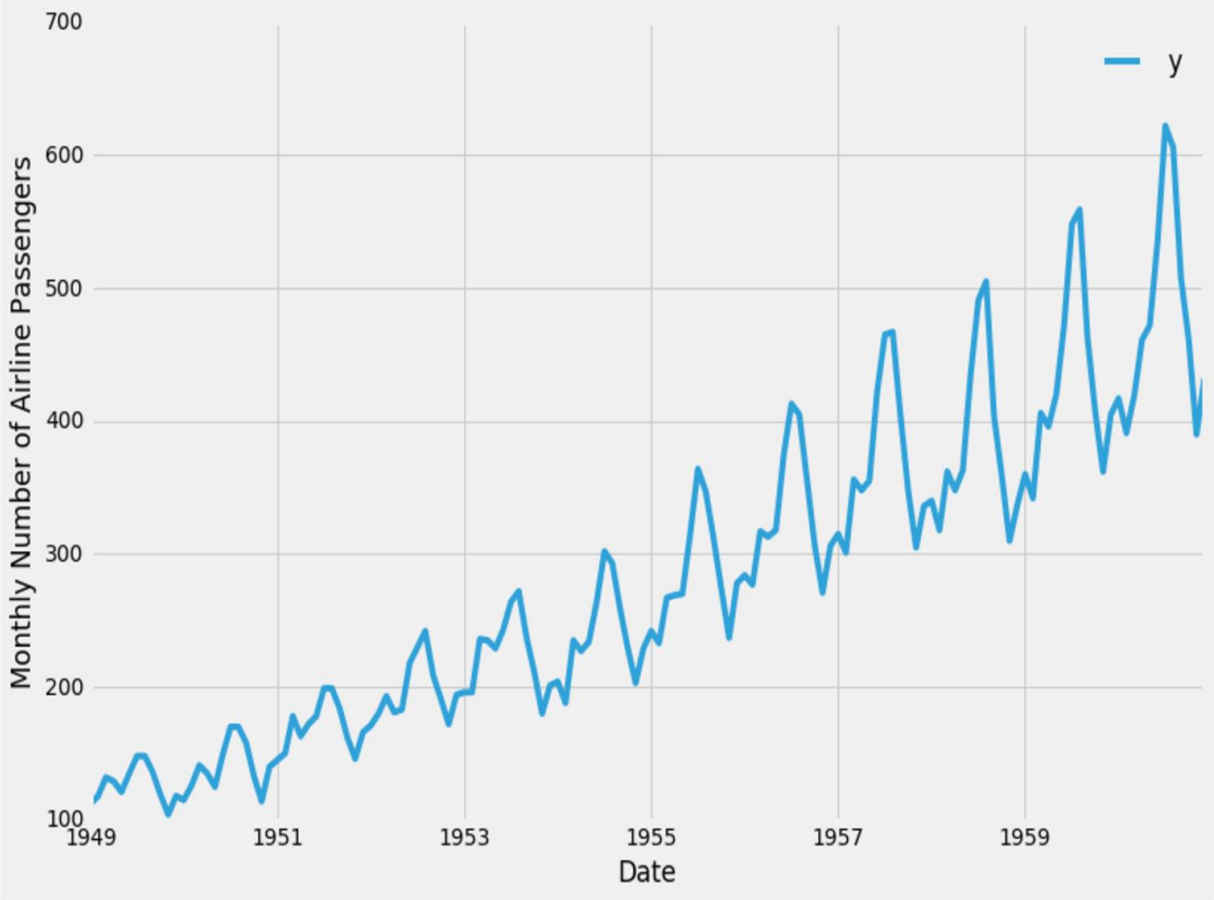
  Time Series Analysis shows a general pattern that upward then it is Upward trend.

- **Downward trend:**

  Time Series Analysis shows a pattern that is downward then it is Downward trend.

- **Horizontal or Stationary trend:**

  If no pattern is observed then it is called a Horizontal or stationary trend.



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Break Even Analysis

Decision Modeling

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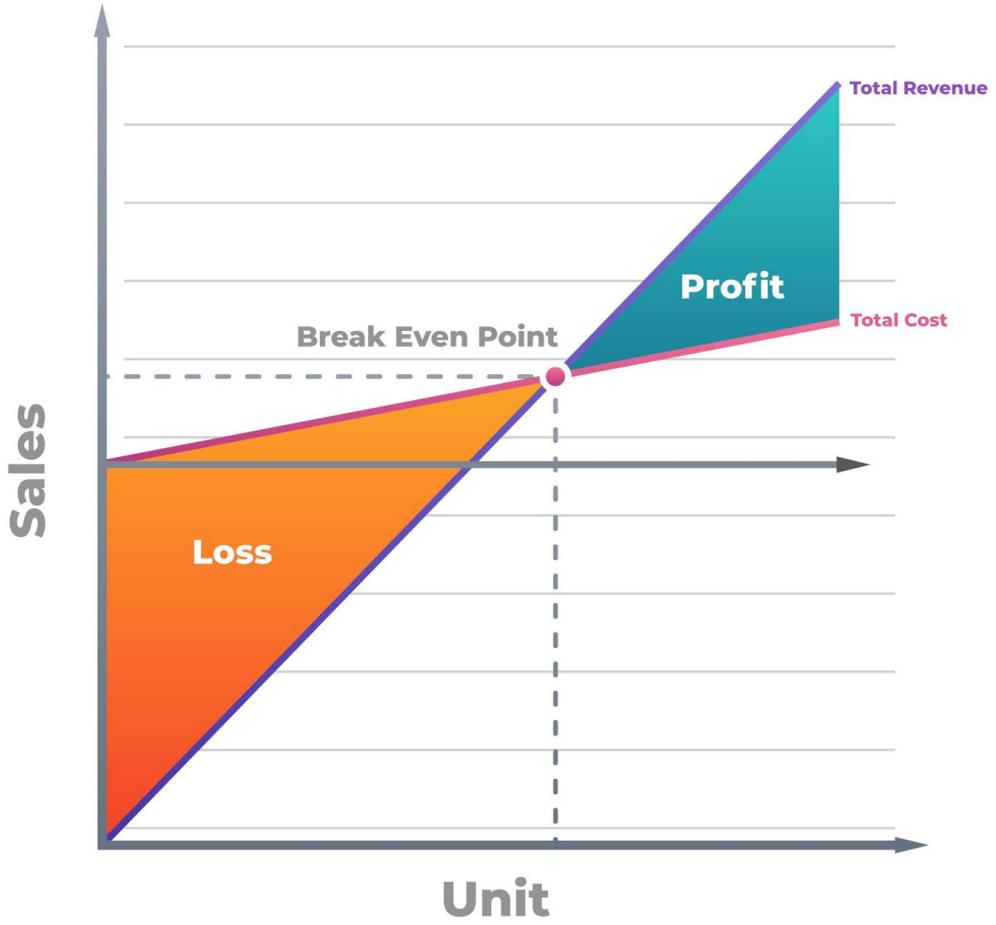
What is Break-even?

Primarily, any venture begun by a firm requires certain capital costs.

It is expected that the business recovers this initial capital through earnings over a period of time.

The position of *no profit - no loss* is termed as Break-Even for a venture.



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Decision Modeling
Software Purchase or In-House?

The business decision needs to be made to make or buy a software product.

- Software development project costs $200,000 and maintenance of the software infrastructure costs $30 per user per annum.
- Buying a SAAS product directly from a software vendor. This has no development or upfront costs, and the software vendor charges $250 per user per annum.

Decision is to choose more economic model based on the number of users.

Assume that development cost of $200k is required to maintain and add new feature every year.

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Break Even Decision Model

- Total Cost of Development = $200,000 + $30 x Users
- Total Cost of SAAS software = $250 x Users

•<u>Breakeven Point</u>:

- Can get break even point by equating above equations

•$200,000 + $30 x Users = $250 x Users

 220xusers = 200,000 about 909 users.

This means at 909 users both costs are same. So, if number of users are more than 909, then we develop software otherwise buying SAAS software is better.

•**General Formula**

•$F + VQ = CQ$

- $Q = F/(C - V)$

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